

Date	2/11/2025
Time id	NM2025TMID07789
Project name	Calculating Family Expenses Using Service Now
Maximum mark	4 marks

PROJECT DESIGN PHASE 2

TECHNOLOGY STACK(ARCHITERTURE& STACK)

Technical Architecture :

The technical architecture of the Family Expense Calculation using ServiceNow project defines how different layers work together to manage and automate expense tracking. The Presentation Layer provides user-friendly dashboards and forms for input and visualization. The Application Layer handles logic and workflows to calculate and categorize expenses. The Data Layer securely stores expense and family information in ServiceNow tables. The Integration Layer enables communication with external systems through APIs. The Security Layer ensures role-based access and authentication. Finally, the Reporting Layer generates analytics and reports for expense insights. Together, these layers create an efficient, automated, and secure expense management system.

TECHNICAL ARCHITECTURE

Family Expense Calculation using ServiceNow

Layer	Components	Technology
Presentation Layer	User Interface (Forms, Dashboards)	ServiceNow UI Builder, HTML, CSS
Application Layer	Expense Calculation Logic & Workflow Automation	ServiceNow Flow Designer, Business Rules
Data Layer	Expense Records, Family Member Data	ServiceNow Tables, CMDB
Integration Layer	Data Import/Export and API Connections	REST API, JSON ServiceNow IntegrationHub
Security Layer	Authentication, Role-based Access Control	ServiceNow ACLs, Single Sign-On (SSO)
Reporting Layer	Expense Summary Reports, Analytics	ServiceNow Performance Analytics, Reporting Engine

Components and Technologies:

Layer	Components	Technologies / Tools
1. Application Layer	Expense Calculation Logic & Workflow Automation	ServiceNow Flow Designer, Business Rules
3. Data Layer	Expense Records, Family Member Data	ServiceNow Tables, CMDB
4. Integration Layer	Data Import/Export and API Connections	REST API, JSON, ServiceNow IntegrationHub
5. Security Layer	Authentication, Role-based Access Control	ServiceNow ACLs, Single Sign-On (SSO)
6. Reporting Layer	Expense Summary Reports, Analytics	ServiceNow Performance Analytics, Reporting Engine